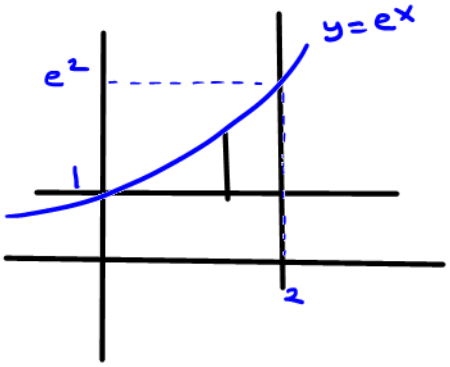
D: $y=1$ $y=e^2$ $x=2$ $x=\ln y$ → $y=e^x$

$\int_1^{e^2}\int_{\ln y}^{2} f(x,y)dxdy$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen yeni integral aşağıdakilerden hangisidir?

A) $\int_0^2\int_1^{e^x} f(x,y)dydx$ B) $\int_{\ln y}^2\int_1^{e^2} f(x,y)dydx$ C) $\int_0^2\int_0^{e^x} f(x,y)dydx$

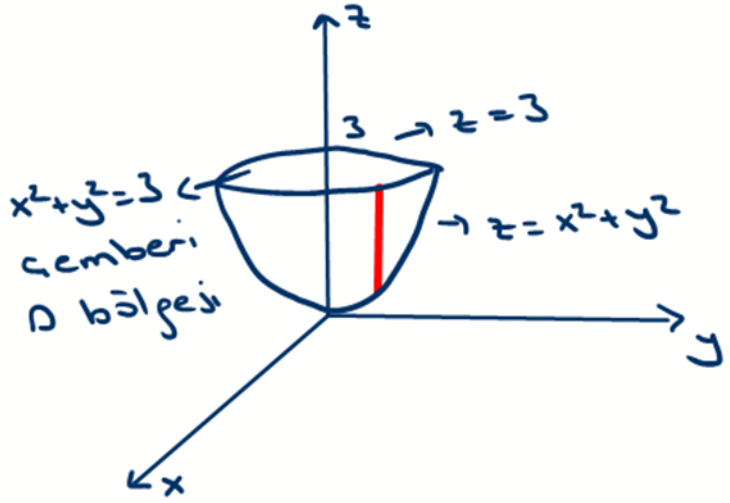
D) $\int_1^{e^2}\int_{e^x}^2 f(x,y)dydx$ E) $\int_1^{e^2}\int_{e^x}^1 f(x,y)dydx$



$$I=\int_0^2\int_1^{e^x} f\,dy\,dx$$

Kolları yukarı, T.N $z=0$ paraboloid

$z=x^2+y^2$ yüzeyi ve $z=3$ düzlemi arasında kalan hacmi veren integral aşağıdakilerden hangisidir?

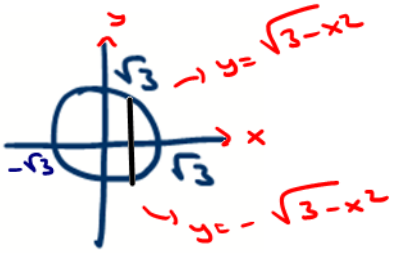


$$V=\iint_D (43k.)\,dx\,dy$$

$$=\iint_D (3-x^2-y^2)\,dx\,dy$$

$$=\int_{-\sqrt3}^{\sqrt3}\int_{-\sqrt{3-x^2}}^{\sqrt{3-x^2}} (3-x^2-y^2)\,dy\,dx$$

D bölgesi:



Kutupsal koor. ya geçelim:

$x=r\cos\theta$

$y=r\sin\theta$

$r^2=x^2+y^2$

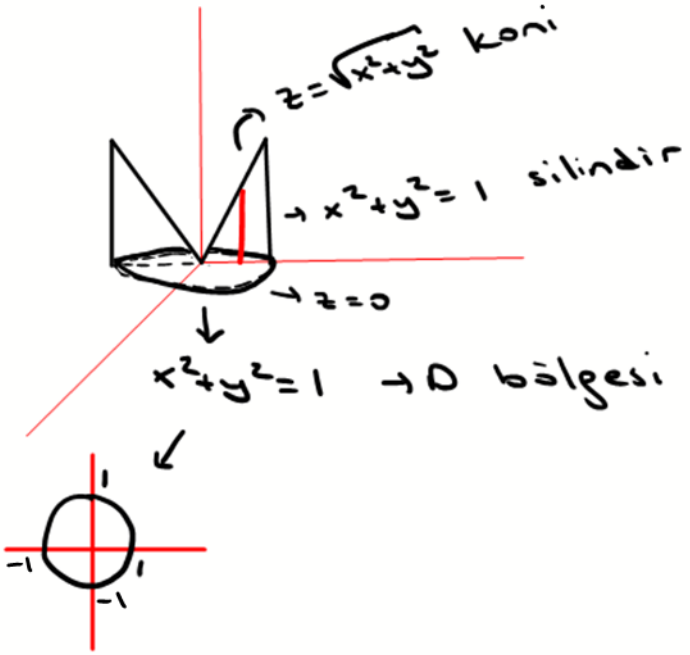
$dx\,dy=r\,dr\,d\theta$

$$V=\int_0^{2\pi}\int_0^{\sqrt3} (3-r^2)\,r\,dr\,d\theta$$



koni

üstten $z=\sqrt{x^2+y^2}$ (silindir) yüzeyi, alttan $z=0$ düzlemi, yandan $x^2+y^2=1$ yüzeyi ile sınırlı cismin hacmini veren integral?



$$V=\iint_D \underbrace{(\text{yüksekli̇k})}_{\sqrt{x^2+y^2}-0}\,dx\,dy$$

$$=\iint_D \sqrt{x^2+y^2}\,dx\,dy$$

$$V=\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{x^2+y^2}\,dy\,dx$$

veya

$$V=\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \sqrt{x^2+y^2}\,dx\,dy$$

Kutupsal koor. yazarsak:

$x=r\cos\theta$

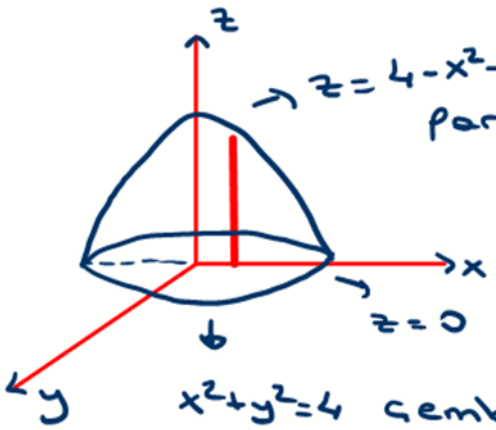
$y=r\sin\theta$

$x^2+y^2=r^2$

$dx\,dy=r\,dr\,d\theta$

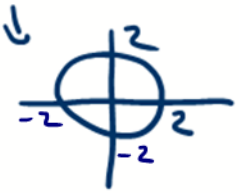
$$V=\int_0^{2\pi}\int_0^1 r\cdot r\,dr\,d\theta$$

$z = 4 - x^2 - y^2$ ve $z = 0$ yüzeylerinin sınırladığı cismin hacmini veren int.?



$x^2 + y^2 = 4$ çemberi

D bölgesi



$$V = \iint_D (4 - x^2 - y^2)\, dx\, dy$$

$$= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - x^2 - y^2)\, dy\, dx$$

veya

$$V = \int_{-2}^{2} \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (4 - x^2 - y^2)\, dx\, dy$$

Kutupsal koor. yazarsak:



$x = r\cos\theta$

$y = r\sin\theta$

$x^2 + y^2 = r^2$

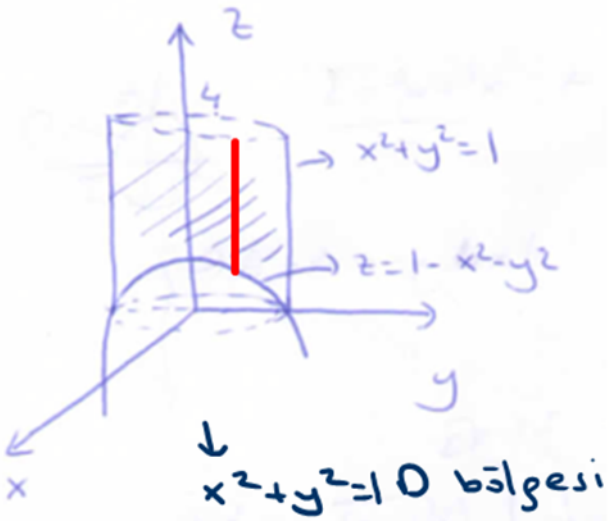
$dx\, dy = r\, dr\, d\theta$

$$V = \iint_D (4 - x^2 - y^2)\, dx\, dy$$

$$= \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr\, d\theta$$

Silindir — Düzlem — Paraboloid

$x^2+y^2=1$, $z=4$, $z=1-x^2-y^2$ yüzeyleri ile sınırlı cismin hacmini veren integral?



$x^2+y^2=1$ bölgesi



$$V=\iint_D \underbrace{(4-k)}_{4-(1-x^2-y^2)}\,dx\,dy$$

$$=\iint_D (3+x^2+y^2)\,dx\,dy$$

$$V=\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (3+x^2+y^2)\,dy\,dx$$

veya

$$V=\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (3+x^2+y^2)\,dx\,dy$$

Kutupsal koor. yazarsak:



$x^2+y^2=r^2$

$dx\,dy=r\,dr\,d\theta$

$x=r\cos\theta$

$y=r\sin\theta$

$$V=\iint_D (3+x^2+y^2)\,dx\,dy$$

$$V=\int_0^{2\pi}\int_0^1 (3+r^2)\,r\,dr\,d\theta$$

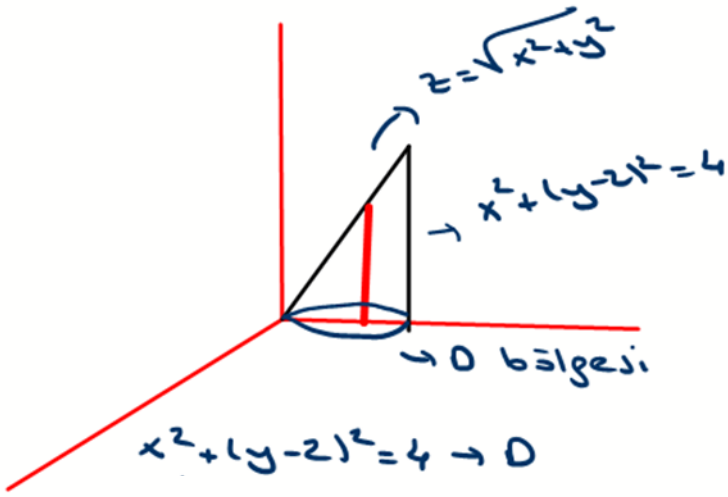
$x^2+y^2-4y=0$, $z=0$, $z=\sqrt{x^2+y^2}$ yüzeyleri arasındaki cismin hacmini veren integral?

(*) $x^2+y^2-4y+4-4=0$

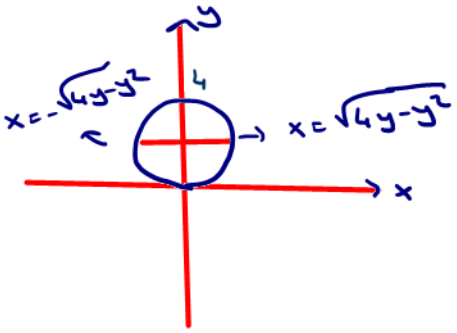
$x^2+(y-2)^2=4$

ötelenmiş silindir

(*) $z=\sqrt{x^2+y^2}$ → Koni



$$V=\iint_D \sqrt{x^2+y^2}\,dx\,dy = \int_0^4 \int_{-\sqrt{4y-y^2}}^{\sqrt{4y-y^2}} \sqrt{x^2+y^2}\,dx\,dy$$



$x^2=4y-y^2$ → $x=\pm\sqrt{4y-y^2}$

Kutupsal koor. yazarsak!



$x=r\cos\theta$

$y=r\sin\theta$

$dx\,dy=r\,dr\,d\theta$

$x^2+y^2=r^2$

$x^2+y^2=4y$ → $r^2=4r\sin\theta$

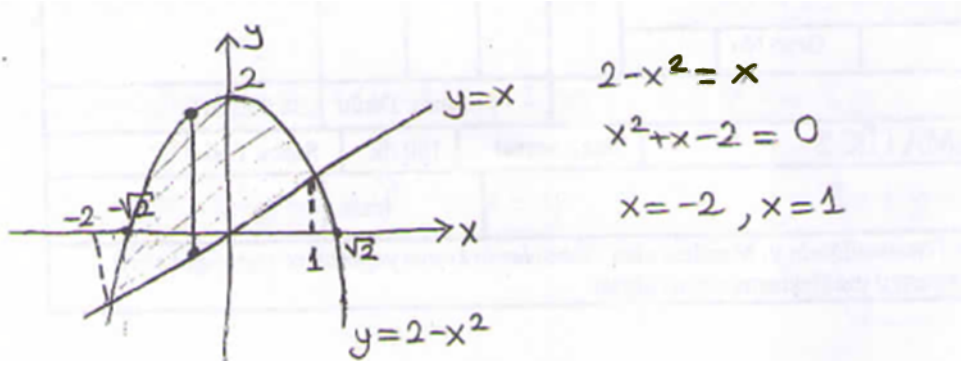
$r=4\sin\theta$

$$V=\iint_D \sqrt{x^2+y^2}\,dx\,dy$$

$$=\int_0^{\pi}\int_0^{4\sin\theta} r\cdot r\,dr\,d\theta$$

D bölgesi $y = 2-x^2$ ve $y = x$ ile sınırlı bölge olsun. $z = 2x^2y$ ile üstten, $z=0$ daki D bölgesi ile alttan sınırlanan cismin hacmini veren integral?

a) $\int_0^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$ b) $\int_{-2}^1 \int_x^{2-x^2} 2x^2 \, dy\,dx$ c) $\int_{-2}^2 \int_0^{2-x^2} 2x^2y \, dy\,dx$



$2-x^2 = x$

$x^2+x-2 = 0$

$x=-2, \; x=1$

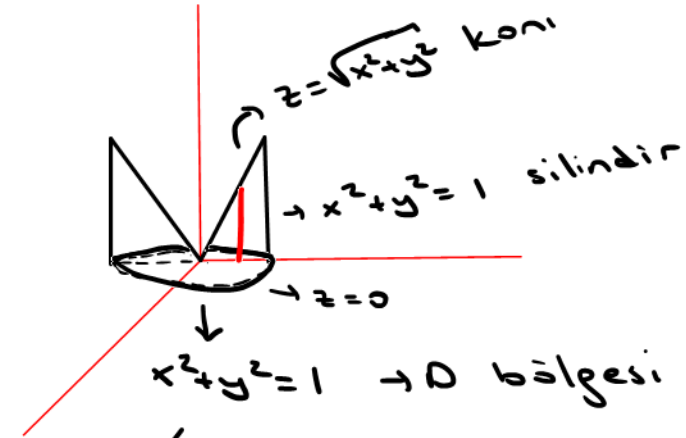
$$V = \iint_D 2x^2y \, dx\,dy$$

$$= \int_{-2}^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$$

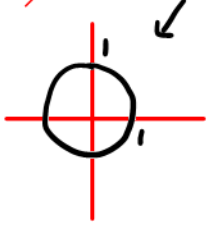
Cevap B

Üstten $z=\sqrt{x^2+y^2}$ yüzeyi, alttan $z=0$ düzlemi, yandan $x^2+y^2=1$ yüzeyi ile sınırlı cismin hacmini veren integral?

a) $\int_0^{2\pi} \int_0^1 r \, dr\,d\theta$ b) $\int_0^{2\pi} \int_0^1 r^2 \, dr\,d\theta$ c) $\int_0^{\pi} \int_0^1 r^2 \, dr\,d\theta$



$x^2+y^2=1 \rightarrow D$ bölgesi



$x = r\cos\theta$

$y = r\sin\theta$

$x^2+y^2 = r^2$

$dx\,dy = r\,dr\,d\theta$

$$V = \iint_D \underbrace{(\text{yükseklik})}_{\sqrt{x^2+y^2}-0} \, dx\,dy$$

$$= \iint_D \sqrt{x^2+y^2} \, dx\,dy$$

$$= \int_0^{2\pi} \int_0^1 r \cdot r \, dr\,d\theta$$

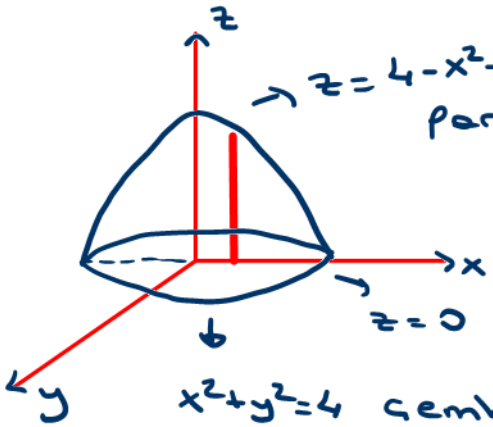
Cevap B

$z = 4 - x^2 - y^2$ ve $z = 0$ yüzeylerinin sınırladığı cismin hacmini veren int.?

a) $\int_0^{2\pi} \int_0^4 r\,dr\,d\theta$

b) $\int_0^{2\pi} \int_0^2 (4r - r^3)\,dr\,d\theta$

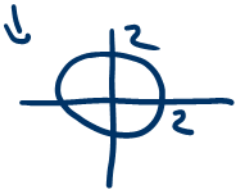
c) $\int_0^{2\pi} \int_0^1 (4r - r^3)\,dr\,d\theta$



$z = 4 - x^2 - y^2$ paraboloid

$x^2 + y^2 = 4$ çemberi

D bölgesi



$$V = \iint_D (4 \cdot 1 \cdot k)\,dx\,dy$$

$$= \iint_D (4 - x^2 - y^2)\,dx\,dy$$

$x = r\cos\theta$

$y = r\sin\theta$

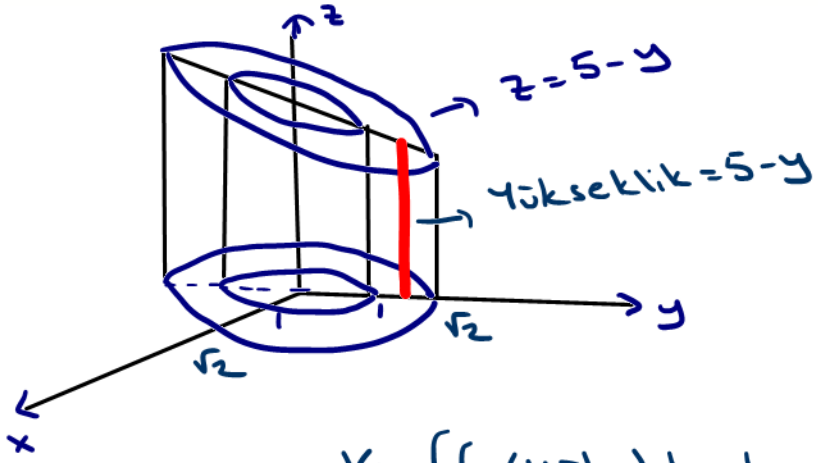
$x^2 + y^2 = r^2$

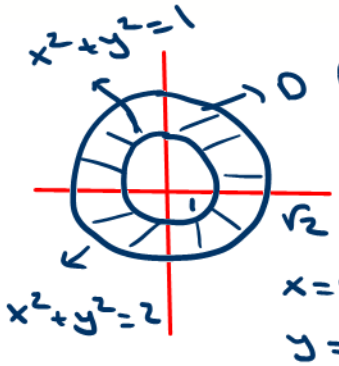
$dx\,dy = r\,dr\,d\theta$

$$= \int_0^{2\pi} \int_0^2 (4 - r^2) r\,dr\,d\theta$$

Cevap B

$z=5-y$ düzleminin altında, $z=0$ düzleminin üstünde, $x^2+y^2=2$ ve $x^2+y^2=1$ silindirleri arasında kalan cismin hacmini veren int.?





$x=r\cos\theta$

$y=r\sin\theta$

$dxdy=rdrd\theta$

$$V=\iint_0 (\text{yük.})\,dxdy$$

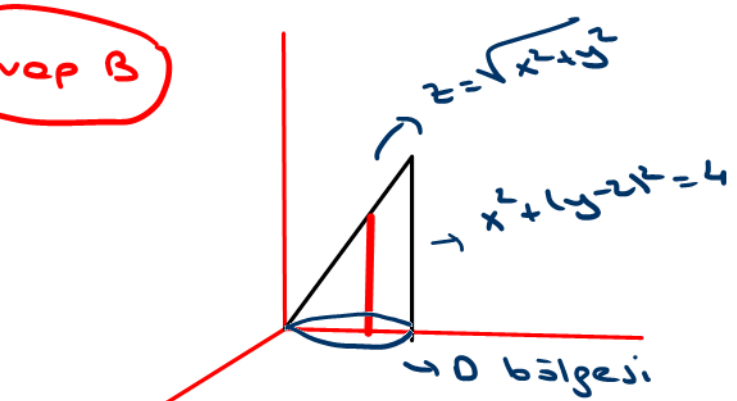
$$=\iint_0 (5-y)\,dxdy=\int_0^{2\pi}\int_1^{\sqrt{2}} (5-r\sin\theta)\,r\,dr\,d\theta$$

Cevap A

$x^2+y^2-4y=0$, $z=0$, $z=\sqrt{x^2+y^2}$ yüzeyleri arasındaki cismin hacmini veren integral?

a) $\int_0^{2\pi}\int_0^{4} r^2\,dr\,d\theta$ b) $\int_0^{\pi}\int_0^{4\sin\theta} r^2\,dr\,d\theta$ c) $\int_0^{\pi}\int_0^{2} r^2\,dr\,d\theta$

Cevap B



(*) $x^2+y^2-4y+4-4=0$

$x^2+(y-2)^2=4$

ötelenmiş silindir

(x) $z=\sqrt{x^2+y^2}$ → Koni

$x^2+(y-2)^2=4 \to 0 \to x^2+y^2=4y \to r^2=4r\sin\theta$

$r=4\sin\theta$



$x=r\cos\theta$

$y=r\sin\theta$

$dxdy=rdrd\theta$

$$V=\iint_0 \sqrt{x^2+y^2}\,dxdy$$

$$=\int_0^{\pi}\int_0^{4\sin\theta} r\cdot r\,dr\,d\theta$$